Exp 15

Exploratory Data Analysis on the Iris Dataset

Aim:

To perform exploratory data analysis on the built-in `iris` dataset by examining dimensions, summary statistics, standard deviation, quantiles, grouping by Species, creating a pivot table, and categorizing Sepal.Length.

Procedure:

1. Load the `iris` dataset.
2. Display its dimensions and summary statistics.
3. Calculate standard deviations and quantiles.
4. Group the data by Species.
5. Create a pivot table.
6. Categorize Sepal.Length into three groups.

Code:

```r
data(iris)

print(dim(iris))
print(summary(iris))

print(sapply(iris[, 1:4], sd))

print(quantile(iris$Sepal.Length))

species_mean <- aggregate(
  . ~ Species,
  data = iris,
  FUN = mean
)

print(species_mean)

pivot_table <- with(
  iris,
  table(Species, Sepal.Width > 3)
)

print(pivot_table)

iris$Sepal_Category <- cut(
  iris$Sepal.Length,
  breaks = c(0, 5, 6, Inf),
  labels = c("Short", "Medium", "Long"),
  right = FALSE
)

print(head(iris[, c("Sepal.Length", "Sepal_Category")]))

print(table(iris$Sepal_Category))
```

OUTPUT:

```
[1] 150   5
  Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
 Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
 1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
 Median :5.800   Median :3.000   Median :4.350   Median :1.300
 Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
 3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
 Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
       Species
 setosa    :50
 versicolor:50
 virginica :50



Sepal.Length  Sepal.Width Petal.Length  Petal.Width
   0.8280661    0.4358663    1.7652982    0.7622377
  0%  25%  50%  75% 100%
 4.3  5.1  5.8  6.4  7.9
     Species Sepal.Length Sepal.Width Petal.Length Petal.Width
1     setosa        5.006       3.428        1.462       0.246
2 versicolor        5.936       2.770        4.260       1.326
3  virginica        6.588       2.974        5.552       2.026

Species      FALSE TRUE
  setosa         8   42
  versicolor    42    8
  virginica     33   17
  Sepal.Length Sepal_Category
1          5.1         Medium
2          4.9          Short
3          4.7          Short
4          4.6          Short
5          5.0         Medium
6          5.4         Medium
```